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PI Name:	Billman, Dorrit Ph.D.		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
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Task Description:	Understanding what properties of skills and knowledge (S&K) support useful generalization and long-term retention is critical to the success of future long-duration missions (e.g., DRM:Mars), yet there is a gap in our understanding of these properties (Space Human Factors Engineering-SHFE-TRAIN-04). Broadly, it is often the case that specialized procedural skills may be well-retained but difficult to generalize and abstract declarative knowledge may support generalization but is difficult to retain. Nevertheless, findings from the literature suggest that integrated knowledge is likely to be both generalizable and well retained. By integrated skills and knowledge we refer to ensembles of a) procedural skills and b) knowledge of principles that are integrated through relations such as explanation, instantiation, and associated use. We will conduct literature research followed by piloting and experimentation to a) define appropriate measures and manipulations and b) assess the extent that integrated S&K generalizes well and is well-retained relative to S&K covering related content but without the integrating relationships. Experimentation will aim to produce integrated knowledge though a variety of learning activities that a) build links from procedural skills to abstract principles through processes such as self-explanation, thus facilitating their decomposition and reuse and b) build links from abstract principles to specific procedures through processes such as instantiation, thus aiding their application. Evidence shows that integrated knowledge is better retained, as the many links of integrated knowledge provide multiple, alternative retrieval cues accessible in many contexts. We will assess whether manipulations that produce integrated knowledge also produce better generalization (as well as measuring retention), relative to knowledge that spans related content but is not integrated, and whether such effects can be attributed to the mediating role of integrated knowledge. We will use the work domain of operating and trouble-shooting complex technology, specifically spacecraft life-support systems as the test case for the proposed research. We will investigate “performing system-related tasks in highly autonomous environments” (Human Exploration Research Opportunities (HERO) Appendix A-25), such as operation, maintenance, and troubleshooting components within habitat life-support systems. Three factors motivate this choice.1) Such work is highly relevant to long-distance, crewed missions. 2) Prior research has studied operation of devices and thus provides findings on representations supporting generalization in similar contexts. 3) We will use an existing software suite including simulation of life-support devices on the International Space Station (ISS) and procedures for their operation as our test bed. This “micro-world” enables experimental control yet is a close analog of real mission work. In this domain the critical skills and knowledge are primarily cognitive (e.g., deciding what procedure to use), rather than sensory-motor (e.g., how hard a wrench must be turned). We include “meta-knowledge” such as identifying what information is missing when resources such as just-in-time-training are relevant. We will assess whether interventions related to integration produce forms of S&K that generalize well and are well retained, a prediction motivated by prior findings but not investigated directly, nor in NASA-relevant context. If integration is the basis for the predicted outcomes, it will provide a powerful principle for identifying and creating forms of S&K that are both generalizable and retainable. If not, we have discovered how learning is affected by important training interventions in a domain highly relevant to NASA future crewed exploration missions. This research will narrow the gap in understanding the factors that make skills and knowledge for NASA-critical tasks more generalizable and more retainable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Technical work across science, technology, engineering, and mathematics (STEM) disciplines, on Earth as in space, requires mastery of complex suites of knowledge and skills. Whether presented as education or as job-specific training, these skills and knowledge must be learned. Much of this work has an open, changing character such that even if time were available, it is not possible to anticipate and teach all the components that will be needed in the work. Thus, training that allows a person effectively to transfer skills and knowledge to situations and tasks that were not trained is extremely valuable. Indeed, transfer to different settings (from training environment to actual work) is often considered the primary success criterion for training; just a transfer from specific problem and procedures used in the classroom to novel situations is a critical goal of education. Understanding what types of knowledge and skills support transfer (and retention) will facilitate more effective training, for technical domains related to those we study.
	We conducted an experiment to assess the effects of alternative training methods on trainee’s ability to generalize their training to new situations and goals. Many technical and scientific domains involve both knowledge of general domain principles, such as those determining how a device works, and skills or methods for getting things done. In working with equipment, it is valuable to learn both about how a device works and about the procedures or methods for using the device. In addition, we claim that learning will be improved by training in which learners actively build relationships between a) the device model with its underlying principles and b) the procedures for operating the device. In particular, we propose that linking an understanding of the device with procedural skills for its operation will improve learners’ ability to generalize. Learners with integrated skills and knowledge of this sort will be better able to generalize their knowledge to new situations. They will be better able to accomplish work with goals, constraints, or resources different from those encountered in training. For example, given training that only covered bringing up and powering down equipment, trainees with integrated training might be better able to generalize to a maintenance goal of testing a component. They might also be better able to generalize to unexpected conditions in which equipment was initially configured differently than had been encountered in training. Because technical work is often complex and unpredictable, it can be impossible or impractical to train all tasks or situations. Thus generalizing to untrained cases is often very valuable. Experiment Prediction. Our experiment compared a training condition emphasizing integration (Integrated Condition) with a training condition that taught the same device model components and the same procedural skill components but did not emphasize linking the two (Component-wise). We predicted enhanced generalization in the Integrated versus Component-wise Condition, particularly as the novel tasks became increasingly different from those in training. An alternative possibility is that separating, rather than integrating, training of these two aspects produces better learning because it places less burden on working memory, and reduces cognitive load to appropriate levels. Method. Our experiment trained users how to operate simulated habitat equipment on the International Space Station, using procedure automation software to carry out procedures. It compared Integrated versus Component-wise Conditions, a

Task Progress:	between-subjects factor. Conditions provided different training and performance on novel, generalization tasks during the transfer phase was assessed. Participants. We recruited students majoring in aero-astronautics engineering. We believed this population of participants would likely be familiar with basic engineering concepts and skills, such as familiarity with schematic representation of devices and with learning complex software. Students were recruited from two universities and ranged from undergraduate to Ph.D. students. The Integrated Condition had 13 participants and the Component-wise had 14. Materials. The experiment used PRIDE procedure software and associated simulation developed by TRAClabs. Both conditions had three main components of the training materials: expository slides, slides presenting questions for the participant to answer orally for the experimenter, and directed hands-on use of the software suite. These materials were closely matched for the two Conditions, except for differences carefully introduced to aid integration. Two key differences were organization of material and the nature of “paper and pencil” study exercises. The Integrated Condition interleaved instruction about the systems to be controlled with instruction about the procedures and procedure software used to operate the systems; it provided exercises that required users to map back and forth between device schematics and operational procedures and others that required reasoning about how device principles constrained the order of procedure steps. The Component-wise Condition blocked instruction by topic and provided study exercises based on identifying elements either in schematics or in procedures and recall rather than inference. Both conditions had the same transfer tasks. The four tasks presented immediately after training all were tasks executing procedures to operate equipment. Task 1 was very similar to activities in training, requiring little generalization. Task 2 differed from training activities in requiring participants to identify and execute parts of a procedure. Tasks 3 and 4 differed in that the conditions required for successful execution were not met, and participants had to reason about what procedures would change these conditions so that the task goals could be achieved. Task 5 differed in the nature of the work: participants were asked to write a procedure for a novel goal of testing a valve. After drafting their procedure they were asked to now execute the valve test. Additional transfer tasks asked users to reason about system states and learn about a new device. Procedure. The experiment was conducted at NASA Ames, was scheduled for 6.5 hours (though sometimes completed in less time) including breaks. Participants worked through the training slides and exercises, self-paced but with experimenter guidance and feedback. The goal was to ensure each participant learned the material well enough to answer the questions posed in the slides and do the hands-on exercises successfully. Following training, participants in both conditions did the identical transfer tasks. Coding. For each execution task we built a coding scheme identifying key behaviors and allowing categorization of accomplishing or failing to accomplish the task goals. We then coded the activity stream from logfiles, to determine task success; for successful trials we compared execution times. Results Broadly, tasks were of appropriate difficulty and participants were engaged in the experiment. Individual difference were large, but patterns of performance on the transfer tasks suggested better generalization in the Integrated than Component-wise Condition. As expected, performance on both success rate and time was similar in both conditions for Task 1. Two tasks required new manipulation of software, either using components of a procedure in Task 2 or designing a new procedure from components in Task 5. Success rate on Task 2 was 92% in the Integrated versus 50% percent in the Component-wise, a reduction in failure to accomplish task goals of 42%. Success rate on Task 5 was 55% in the Integrated versus 22 % percent in the Component-wise, a reduction in failure to accomplish task goals of 33%. Execution times were similar between conditions. Tasks 3&4 required reasoning about how to identify what procedures should be run in order to prevent procedure failures and correct the conditions needed for procedures named in the task. Execution times on successful trials were 7m50sec and 7m07sec in the Integrated Condition but 12m02sec and 9m40sec in the Component-wise Condition. Discussion This study found suggestive evidence of better generalization following training that emphasized integration. The bulk of training was the same in both condition; the same information and activities were provided about the systems to be controlled and about the software and procedures for controlling them. The differences introduced to foster integration were relatively modest. Nevertheless, performance patterns showed in increased success rate and decreased times for the Integrated Condition. Future work is needed to replicate and extend these findings, with larger numbers of participants and perhaps participants with more homogeneous initial knowledge as indexed by current education level. We also plan to use an initial, engineering knowledge assessment. In addition, we will assess retention over periods of weeks of these skills and knowledge including the ability to flexibly apply them to novel situations.
Bibliography Type:	Description: (Last Updated: 10/03/2019)
Abstracts for Journals and Proceedings	Billman D, Schreckenghost D, Caddick Z. "Transfer at the Level of Human-Computer System: Problem Solving using Procedure-Automation Software." COGSCI 2016, 38th Annual Conference of the Cognitive Science Society, Philadelphia, PA, August 10-13, 2016. In: Proceedings of the 38th Annual Conference of the Cognitive Science Society (Papafragou, A., Grodner, D., Mirman, D., & Trueswell, J.C. (Eds.). Austin, TX: Cognitive Science Society, 2016. , Aug-2016